Automated LPG Safety System with "Real-Time Alerts and Absorption Control"

PADMANABAN R
Assistant Professor, SCOPE
Vellore Institute of Technology, CHENNAI
Chennai, India
padmanaban.r@vit.ac.in

SURYA M S
MTech Integrated Computer Science and Business Analytics
Vellore Institute of Technology, CHENNAI
Chennai, India
surya.ms2022@vitstudent.ac.in

AARMITHA A
MTech Integrated Computer Science and Business Analytics
Vellore Institute of Technology, CHENNAI
Chennai, India
aarmitha.a2022@vitstudent.ac.in

***Abstract*—The Automated LPG Safety System with Real-Time Alerts and Absorption Control is intended to improve safety in domestic and industrial settings through LPG gas leak detection, real-time alerts, and automatic control of gas absorption. The system utilizes gas sensors (MQ-6/MQ-5) to read LPG levels continuously and incorporates an MQ-135 sensor to determine the saturation of the absorbing material. There is a specialized compartment with an inexpensive mixture of zeolites and activated charcoal. It is activated through a DC motor, solenoid, or servo motor when a leak is detected to trap and absorb the leaked gas. An Arduino Uno is used as a processor, coordinating the collection of sensor data, making decisions, and controlling the actuation process. To provide real-time notifications, the system uses a variety of communication modules: ESP8266/ESP32 for Wi-Fi-based notifications, GSM modules (SIM800L/SIM900A) for SMS notifications, and Bluetooth modules (HC-05/HC-06) for close- range control. A weight sensor (along with an HX711 module) also keeps track of the status of the absorbent material, initiating maintenance notifications upon replacement requirement. This combined method delivers a solid, scalable, and affordable solution to counteract LPG leakage hazards, ensuring intervention in real-time and improving safety.**

Keywords— LPG Leak Detection, Real-Time Gas Monitoring, IoT-Based Safety System, Automated Gas Absorption, Gas Sensors (MQ-6, MQ-5, MQ-135), Activated Charcoal and Zeolite Absorption, Actuation Mechanisms (DC Motor, Servo Motor, Solenoid), Weight Sensor (HX711 Module), Cloud-Based Notification System

I. Introduction

Liquefied Petroleum Gas (LPG) is a fundamental source of energy for domestic and industrial use due to its convenience and cost-effectiveness. Leaks in LPG, however, are dangerous, with risks of fire hazards, explosion, and toxic poisoning. The traditional gas leak detection methods tend to rely on manual check or alarm notification, which will not always be timely and effective in the detection of leaks. The device employs MQ-6 and MQ-5 gas sensors to continuously monitor levels of LPG. An additional MQ-135 sensor checks the saturation of absorbent material to effectively absorb gas. When there is a leak, a compartment containing an affordable mixture of zeolites and activated charcoal is driven by a servo motor, solenoid, or DC motor to suck up the leaked gas. Arduino Uno oversees the whole process by analyzing sensor data, making real-time decisions, and controlling the elements required in a way that it contains the leak.

II. Problem Statement

Liquefied Petroleum gasoline (LPG) is extensively used in residential and industrial settings due to its performance and simplicity of use. but LPG is extraordinarily flammable, or even a small leakage can lead to dangerous situations which include fires, explosions, or toxic exposures. Traditional detection systems often depend upon manual intervention or lack the capability for instant corrective motion, that may bring about delayed responses and multiplied dangers. The absence of an included, automatic gadget that no longer most effectively detects LPG leaks but also takes remedial actions and gives actual-time alerts has created a substantial hole in safety measures.

III. Project Goals

The key objectives of the Automated LPG assurance System with In-time Warning & Absorption Control are:

- **Quick Detection:** Use reliable gas sensors (MQ-6/MQ-5) for continuous monitoring of LPG levels to detect any leak instantly
- **Automated Process:** Turn on an absorption process by a separate compartment containing a blend of activated carbon and zeolites to catch leaked gas.
- **Real-time Alerts:** Take advantage of several communication modules (WiFi, GSM, and Bluetooth) to send instant alerts to users, maximizing responsiveness whenever users are not present on the site.
- **Absorbent Monitoring:** Have the MQ-135 and Weight Sensor (HX711) that could track the saturation status of the absorbent material which could notify you earlier if it needs some maintenance and replacement.
- **User Interaction:** Offer user-friendly interface with manual override features (push button/switch) for reset and system operation control.

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

IV. Significance of the Project

The importance of the development of this automatic LPG safety system can be explained on several grounds:

Improved Safety: By providing early warning and automatic intervention, the system decreases the likelihood of huge accidents and, hence, guard lives and property.

Fast Response: Real-time notifications enable that the correct people or emergency help can be alerted quickly, minimizing the chance for an LPG leakage.

Economical Solution: Components are basic and materials available cheap (activated charcoal and zeolites), this system has an accessible choice for residential and small-scale industry applications.

Herc lability: The craftsman design galvanizes center life enhancing utilization and worldwide community being affirm and changes the customizable panel boards; preserve biostability and time-adaptable for various application and concerning measurements.

Preventive Maintenance: Real-time monitoring of the absorbent material conditions to help in scheduling of maintenance to ensure long term reliability and smooth operating of system.

V. Scope of the Project

This project is the design, build and test of an automated LPG safety prototype. The scope includes:

Hardware Integration: Connection of mandatory hardware components including gas sensor (MQ-6/MQ-5, MQ-135), microcontroller (Arduino Uno), actuators (servo motor/solenoid/DC motor), communication module (ESP8266/ESP32, GSM, Bluetooth) and auxiliary peripherals (breadboard, jumper wires, push button).

Program software development: Programming the microcontroller to interpret data from the sensor, operate the logic for control and manage communication protocols for real-time notifications.

System Testing: Laboratory simulations with sensor to verify calibration, response time, reliability of alarm, performance of absorption control.

Documents: System implementations Full documentation regarding experiment results Challenges faced during the project Future advancement prospects.

VI. UNITS AND EQUATIONS

A. *Abbreviations and Acronyms*

LPG – Liquefied Petroleum Gas
PPM – Parts Per Million
MQ-6/MQ-5 – Gas Sensors for LPG Detection
ESP8266/ESP32 – Wi-Fi Communication Modules
GSM – Global System for Mobile Communications
HC-05/HC-06 – Bluetooth Modules for Wireless Communication
DC Motor – Direct Current Motor
HX711 – Load Cell Amplifier Module for Weight Measurement

B. *Units*

The relevant units for your Automated LPG Safety System with Real-Time Alerts and Absorption Control are:

- **Gas Concentration:** Parts Per Million (**PPM**)
- **Weight Measurement:** Grams (**g**) or Kilograms (**kg**)
- **Force Measurement:** Newton (**N**)
- **Torque:** Newton-meter (**Nm**)
- **Bluetooth Signal Frequency:** Megahertz (**MHz**) or Gigahertz (**GHz**)
- **Power Consumption:** Watts (**W**)
- **Voltage:** Volts (**V**)
- **Current:** Amperes (**A**)
- **Time (Response Time):** Seconds (**s**)
- **Distance (Signal Transmission Range):** Meters (**m**)

C. *Equations*

1) Gas Concentration Measurement

- **Unit:** Parts Per Million (**PPM**)
- **Equation:** The output of the **MQ-6/MQ-5** sensor follows a logarithmic response

$$R_s = \frac{V_{R_L} \times (R_L)}{(V_{CC} - V_{RL})}$$

R_s = Sensor resistance in the presence of LPG

V_{RL} = Voltage across the load resistor

R_L = Load resistor

V_{CC} = Supply voltage

The **LPG concentration in PPM** is derived from the sensor calibration curve

$$PPM = A \times \left(\frac{R_s}{R_0}\right)^B$$

where A and B are sensor-specific constants, and R0 is the resistance in clean air.

2) Weight Sensor (HX711 Module) for Absorbent Material Monitoring

- **Unit:** Grams (g) or Kilograms (kg)
- **Equation:**

$$Weight = Load\ Cell\ Output \times \quad Reference\frac{Volatge}{Gain}$$

The HX711 module amplifies the analog signal and converts it to digital data.

3) Force and Torque for Compartment Opening Mechanism

- **Unit:** Newton (N) for force, Newton-meter (Nm) for torque
- **Equation for Torque (if using a Servo/DC motor):**

$$T = F \times r$$

where

- T = Torque (Nm)
- F = Force applied (N)
- r = Distance from pivot (m)

4) Power Consumption of Components

a. **Unit:** Watts (W)

b. **Equation:**

$$P = V \times I$$

where:

c. P = Power consumption (W)

d. V = Voltage (V)

- I = Current (A)

5) Response Time of the System

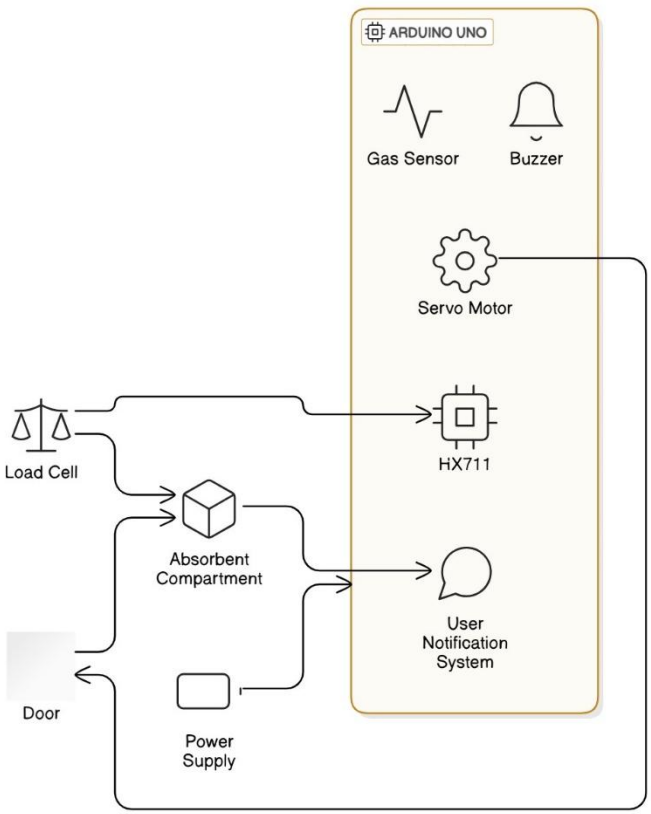
- **Unit:** Seconds (s)
- **Equation:**

$$t_{\text{response}} = t_{\text{sensor}} + t_{\text{actuation}} + t_{\text{notification}}$$

where:

- t_{sensor} = Time taken to detect LPG leakage
- $t_{\text{actuation}}$ = Time taken to open the absorption compartment
- $t_{\text{notification}}$ = Time taken for Bluetooth alert transmission

ARCHITECTURE DIAGRAM



VII. EXPLANATION OF ARCHITECTURE

*The provided **LPG Control Safety Device Flowchart** represents the logical workflow of the automated LPG leakage detection and absorption control system. Below is a breakdown of each step in architecture:*

1. Detection of LPG Leakage

*The system continuously monitors the environment using an **MQ-6 or MQ-5 gas sensor**, which detects the presence of LPG in the air.*

If no LPG is detected, the system remains in the monitoring state.

2. Alert Mechanism

*When LPG leakage is detected, the system **immediately notifies the user** through an IoT-based alert system.*

*This notification is sent via **Wi-Fi (ESP8266/ESP32)** or **GSM Module (SIM800L/SIM900A)**, ensuring that alerts are received even if internet connectivity is unavailable.*

3. Activation of Absorption Mechanism

*The system automatically opens a **compartment containing absorbent material** (activated charcoal + zeolites) to control the leakage.*

*A **servo motor, solenoid, or DC motor** is used to open the compartment, allowing the absorbent material to neutralize the leaked gas.*

4. Continuous Monitoring of Absorption Efficiency

*The system continuously evaluates whether the absorbent material is still effective using an **MQ-6 sensor** (which detects gas saturation levels).*

*Additionally, an **HX711 weight sensor** may be used to track the depletion of the absorbent material.*

5. Material Replacement Notification

*If the absorbent material becomes **saturated or ineffective**, the system **notifies the user** to replace it.*

The system waits for user intervention while continuing to absorb any remaining LPG.

6. User Intervention & System Reset

*Once the user arrives, they manually **confirm control** using a **push button or switch** to deactivate the emergency absorption system.*

*The compartment **closes automatically**, and the system **resets to its original monitoring state**, ensuring continuous safety.*

VIII. SYSTEM REQUIREMENTS

1. Hardware Requirements

A. Core Components

Arduino Uno – Serves as the central microcontroller for processing sensor data and system control.

Breadboard & Jumper Wires – Used for connecting various electronic components efficiently.

Push Button / Switch – Allows manual system to reset or emergency control.

B. Sensing Modules

MQ-6 or MQ-5 Gas Sensor – Detects LPG leakage.

MQ-135 Sensor – Monitors gas absorption efficiency and detects saturation of the absorbent material.

Weight Sensor (HX711) – Measures the weight of absorbent material to determine when it needs replacement.

C. Actuators and Alerts

Servo Motor / Solenoid / DC Motor – Controls the compartment that houses the absorbent material.

Buzzer / Alarm – Provides an audible alert when LPG leakage is detected.

LED Indicators – Signals different system statuses such as normal operation, leakage detected, or material change required.

2. Software Requirements

Arduino IDE – Used for writing and uploading code to the microcontroller.

Embedded C / C++ – Programming languages for system logic and sensor integration.

IoT Platforms (Optional) – Possible cloud-based monitoring platforms such as Blynk, Adafruit IO, or Firebase for remote tracking and control.

3. Communication Requirements

A. IoT & Wireless Modules

ESP8266 / ESP32 Wi-Fi Module – Enables real-time notifications via the internet.

Bluetooth Module (HC-05/HC-06) – Facilitates short-range control via a mobile app.

B. Mobile & Remote Control

Mobile Notification System – Sends alerts for gas leakage, absorbent material replacement, and system status.

Manual Override via Mobile or Button – Allows user intervention in case of emergency.

4. Absorption & Safety Materials

Best Absorbent Combination: Activated Charcoal + Zeolites – Chosen for its cost-effectiveness and efficiency in LPG absorption.

5. Power Requirements

Power Supply (5V or 12V, Depending on Components) – Ensures the proper functioning of

Arduino, sensors, motors, and communication modules.

Battery Backup (Optional, for Uninterrupted Operation)
Provides continuous functionality during power outages.

IX. LITERATURE SURVEY

(1) Embedded-Based LPG Gas Detection and Safety System

This paper deals with an embedded gas leakage sensing system of LPG using Arduino Uno microcontroller and a gas sensor. The system also comes with an automated relay mechanism to regulate gas flow and reacts right away to potential safety threats. The study shows that embedded systems are crucial for efficient safety solutions.

(2) An Intelligent LPG Leakage Detection and Control System

This paper presents an intelligent system which detects LPG gas leakage and cuts off the gas supply automatically from preventing the accidents. The research is a study on sensor-based detection methods and micro controller integration that aims at safety enhancement. The methodology guarantees proactive response, minimizing the hazards arising out of the LPG leak.

(3) Liquefied Petroleum Gas (LPG), Leakage Detection and Monitoring System

An IoT enabled method for LPG leak detection and assessment has been proposed in this. The research in the field of rural housing combines them with wireless sensor networks (WSN) for providing the persons real time alerts and notifications. The system ensures that safety is better facilitated with an immediacy offered by gas leak awareness that doesn't necessarily keep the user within home premises.

(4) LPG Gas Leakage Detection and Monitoring System

The study shows the development of smart gas leak detection system which can be used in residential and industrial areas. The system is based on gas sensors that at detecting leaks sends through messages to your mobile phone. The study shows the capability of systematic gas monitoring for the prevention of accidents.

(5) Smart LPG Leakage Monitoring and Control System

Using Gas Sensor MQ-X, AWS IoT, and ESP Module

This work possesses an improved LPG leakage detecting system based on MQ X gas sensor, AWS IoT, with ESP module. The system will detect gas leaks and send notifications to users through cloud-based alerts in addition to regulating the gas supply. The report highlights the use of IoT to reduce the risk.

(6) Micro gas sensor array with neural network for recognizing combustible leakage gases

A compact gas sensor array was designed using porous tin oxide films combined with noble metal catalysts. These films were integrated onto a micro-hotplate optimized for low power usage, fast response, and even heat distribution. The system was tested for detecting gases like propane, butane, LPG, and carbon monoxide. A multilayer neural network with backpropagation was employed to classify the gas types and their concentrations. Results showed high accuracy in identifying gases at levels below safety thresholds. This highlights the effectiveness of merging sensor arrays with neural networks for smart gas leakage detection.

(7) An Enhanced Smart Intelligent Detecting and Alerting System for Industrial Gas Leakage using IoT in Sensor Network

Gas leakage poses serious safety risks in industrial and domestic environments. This paper presents an IoT-based intelligent system for real-time gas leakage detection, monitoring, and alerting. The system integrates sensors to detect leaks, flames, and usage, and automatically shuts off gas supply to prevent accidents. Alerts are sent via SMS and GPRS, providing users with location data for prompt response. A buzzer is triggered upon fire detection, ensuring local warning. Data is stored on the cloud for tracking daily gas usage and analyzing user consumption. This smart system significantly enhances safety and reduces unnecessary gas waste.

(8) Development of LPG Leakage Detection Alert and Auto Exhaust System using IoT

This paper presents ensuring safety in homes and workplaces is crucial, especially regarding LPG gas leakage. This project introduces an IoT-based system that detects LPG leaks and responds automatically to prevent hazards. Upon detecting gas leakage, the system activates an exhaust fan, shuts off the gas supply via a solenoid valve, and alerts users through a mobile app. A gas sensor highly sensitive to propane and butane enables accurate detection. An LCD displays real-time alerts for local awareness. Additionally, a load cell monitors cylinder weight and allows users to schedule refills. This automated system ensures faster, more reliable responses than traditional manual methods.

(9) IoT-based Gas Leak Detection Device

LPG is widely used in households as a cleaner alternative to kerosene, following national energy conversion efforts. Despite its benefits, LPG is highly flammable and leaks due to faulty installations or damaged cylinders pose serious safety risks. This research presents an IoT-based device using sensors and NodeMCU for continuous gas leak monitoring. The system detects gas presence and immediately sends alerts to users through an Android smartphone. Real-time notifications help users respond quickly and prevent potential accidents. The device offers a cost-effective and efficient safety solution. Its implementation significantly enhances domestic gas safety.

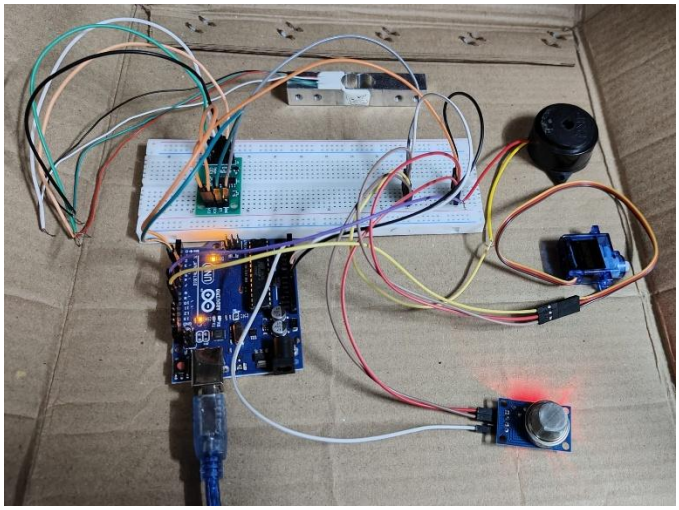
(10) Design and Implementation of an IoT-Based LPG and CO Gases Monitoring System

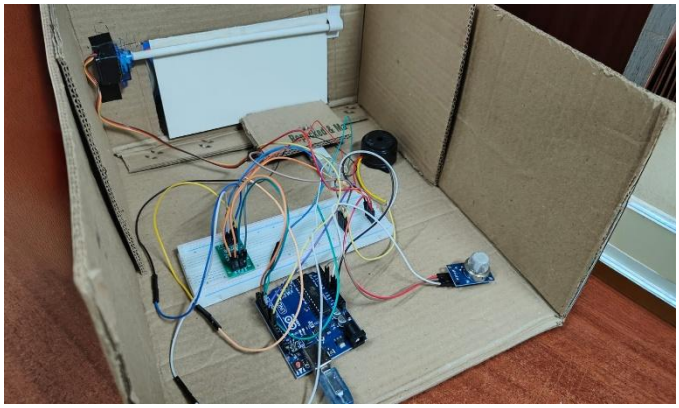
This paper will focus on an LPG and CO gas concentration telemetry Monitoring System using internet of things (IoT). The system gives actual time tracking and alerts, guaranteeing user consciousness of probable CH hazards. The study shows the way to fit IoT with normal gas detection methods for higher safety. The widespread use of LPG, a flammable and asphyxiating gas, along with the dangers of carbon monoxide (CO) exposure, has raised serious health and safety concerns. This study introduces an IoT-based telemetry system for real-time monitoring of LPG and CO gas concentrations. It utilizes a microcontroller and respective sensors to detect gas levels in the environment. Data is transmitted to the cloud via Amazon Web Services (AWS), enabling remote access from any internet-connected device.

`

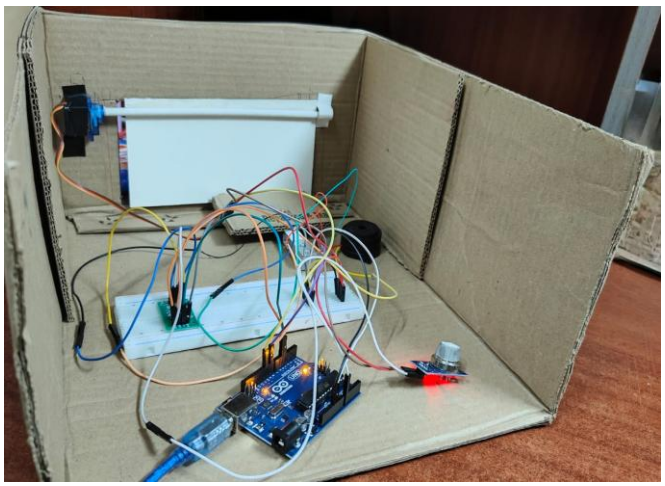
X. EXPERIMENTAL RESULTS

1) HARDWARE:

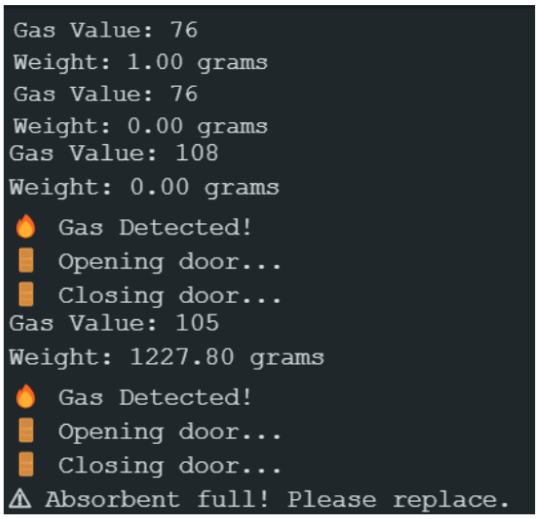




FINAL WORKING PART:



2) SOFTWARE:



```
Gas Value: 76
Weight: 1.00 grams
Gas Value: 76
Weight: 0.00 grams
Gas Value: 108
Weight: 0.00 grams
🔥 Gas Detected!
🚪 Opening door...
🚪 Closing door...
Gas Value: 105
Weight: 1227.80 grams
🔥 Gas Detected!
🚪 Opening door...
🚪 Closing door...
⚠ Absorbent full! Please replace.
```

TABLE FORMAT:

Step	Gas Value	Weight (grams)	Event / Action
1	76	1.00	Gas level normal. No action taken.
2	76	0.00	Gas level normal. No action taken.
3	108	0.00	Gas Detected → Door opened and closed.
4	105	1227.80	Gas Detected → Door opened and closed.
5	—	1227.80	⚠ Absorbent full! Please replace.

EXPLANATION PART:

The experimental output confirms that the LPG leakage detection and safety response system is working as intended. Initially, gas values were below the detection threshold of 100, indicating a safe environment. Once the gas concentration rose to 108 and 105 (values > 100), the system accurately identified this as a hazardous condition and triggered the safety mechanism.

Upon detection:

- The buzzer alerts the user,
- The servo motor opens the sliding door containing the absorbent material,
- The absorbent weight is continuously monitored by a load cell.

As gas exposure occurred, the absorbent's weight increased significantly, eventually reaching 1227.80 grams. This exceeds the defined threshold of 1000 grams for absorbent capacity, triggering the alert:

"Absorbent full! Please replace."

Threshold Summary:

Parameter	Threshold Value	Action Taken
Gas Detection Value	> 100	Trigger buzzer and sliding door mechanism.
Absorbent Weight	> 1000 grams	Notify user to replace absorbent ("Absorbent full").

CONCLUSION:

The experimental results validate the effectiveness of the proposed LPG leakage detection and safety response system. The system reliably identifies hazardous gas levels exceeding the threshold of 100 and promptly initiates safety actions, including activating an alert and opening the absorbent chamber. Real-time monitoring of the absorbent weight ensures continued protection, with a clear notification when the absorbent reaches its full capacity. This integrated approach enhances safety by providing timely alerts and responses, making it suitable for practical deployment in environments prone to LPG leakage.

X. REFERENCES

P. Anandan, Vysyaraju Tharun Raj, Putchakayala Namratha, Abdul Hameed, Balaveerannagari Mahammad Farooq; Embedded based smart LPG gas detection and safety management system. *AIP Conf. Proc.* 3 October 2022
DOI: https://doi.org/10.1063/5.0110382 **(1)**

Prasanta Pratim Bairagi, Bikram Patir, Pranjal Bora, Latika Deka, Sisir Kumar Rajbongshi, "SmartGuard: An Intelligent LPG Leakage Detection and Control System," *International Journal of Engineering Trends and Technology*, vol. 72, no. 9, pp. 265-272, 2024
DOI: https://doi.org/10.14445/22315381/IJETT-V72I9P121 **(2)**

Sitan, Tamil & Ghafar, Aimi. (2018). Liquefied Petroleum Gas (LPG) Leakage Detection and Monitoring System. Journal of Science and Technology.
DOI: 10.30880/jst.2018.10.03.007 **(3)**

G. N. Sai, K. P. Sai, K. Ajay and P. Nuthakki, "Smart LPG Gas Leakage Detection and Monitoring System," *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, Tirunelveli, India, 2023
DOI: 10.1109/ICSSIT55814.2023.10060970 **(4)**

Chakraborty, S. & Aithal, P. S. (2024). Smart LPG Leakage Monitoring and Control System Using Gas Sensor (MQ-X), AWS IoT, and ESP Module. International Journal of Applied Engineering and Management Letters (IJAEML), 8(1), 101-109.
DOI: 10.5281/zenodo.10718875 **(5)**

Dae-Sik Lee, Sang-Woo Ban, Minho Lee and Duk-Dong Lee, "Micro gas sensor array with neural network for recognizing combustible leakage gases," in *IEEE Sensors Journal*, vol. 5, no. 3, pp. 530-536, June 2005,
DOI: 10.1109/JSEN.2005.845186 **(6)**

G. A. Senthil, P. Suganthi, R. Prabha, M. Madhumathi, S. Prabhu and S. Sridevi, "An Enhanced Smart Intelligent Detecting and Alerting System for Industrial Gas Leakage using IoT in Sensor Network," *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, Tirunelveli, India, 2023
DOI: 10.1109/ICSSIT55814.2023.10060907 **(7)**

K. Gavaskar, D. Malathi, G. Ravivarma and A. Arulmurugan, "Development of LPG Leakage Detection Alert and Auto Exhaust System using IoT," *2021 7th International Conference on Electrical Energy Systems (ICEES)*, Chennai, India, 2021
DOI: 10.1109/ICEES51510.2021.9383633 **(8)**

M. Santiputri and M. Tio, "IoT-based Gas Leak Detection Device," *2018 International Conference on Applied Engineering (ICAE)*, Batam, Indonesia, 2018
DOI: 10.1109/INCAE.2018.8579396 **(9)**

David C. Wyld et al. (Eds): BIoT, DKMP, CCSEA, EMSA - 2021 pp. 31-39, 2021. CS & IT - CSCP 2021
DOI: https://doi.org/10.48550/arXiv.2107.07406 **(10)**